DIY e-Handpan Assembly Instructions

Benoit Collin, Dominique Fourer and Eric Genotelle

August 6, 2026

Contents

Workshop Resources **3**

1 Drilling the Bowls **3**
1.1 Required tools . . . 3
1.2 Materials . . . 3
1.3 Procedure . . . 3
1.3.1 Preparing the templates . . . 3
1.3.2 Marking the bowls . . . 3

2 Cutting and Piercing the Foam Discs **4**
2.1 Required tools . . . 4
2.2 Materials . . . 4
2.3 Procedure . . . 4
2.3.1 Drawing the circles on the foam . . . 4
2.3.2 Cutting and piercing the discs . . . 4

3 Attaching the Sensor and Foam Disc to the CD **4**
3.1 Required tools . . . 4
3.2 Materials . . . 4
3.3 Procedure . . . 5
3.3.1 Making the “straw” tool . . . 5
3.3.2 Applying adhesive to the CD . . . 5
3.3.3 Attaching the sensor and foam disc . . . 5

4 Attaching the Striking Surfaces (Pads) to the Bowls **5**
4.1 Required tools . . . 5
4.2 Materials . . . 5
4.3 Procedure . . . 6

5 Making and Attaching the Supports **6**
5.1 Required tools . . . 6
5.2 Materials . . . 6
5.3 Procedure . . . 6

6 Drilling the Bowls for the LEDs **6**
6.1 Required tools . . . 7
6.2 Materials . . . 7
6.3 Procedure . . . 7

7 Inserting the LEDs into the Bowls **7**
7.1 Required tools . . . 7

7.2 Materials . . . 7
7.3 Procedure . . . 7

8 Ground Circuit **7**
8.1 Required tools . . . 7
8.2 Materials . . . 8
8.3 Procedure . . . 8
8.3.1 Connecting the sensor grounds to the LEDs . . . 8
8.3.2 Connecting the LEDs together . . . 8

9 Preparing the DB25 Cables **8**
9.1 Required tools . . . 9
9.2 Materials . . . 9
9.3 Procedure . . . 9
9.3.1 Preparing the identification labels . . . 9
9.3.2 Cutting and stripping the cables . . . 9
9.3.3 Identifying the conductors . . . 9

10 Connecting the DB25 Cables **9**
10.1 Required tools . . . 9
10.2 Materials . . . 10
10.3 Procedure . . . 10
10.3.1 Preparation . . . 10
10.3.2 Connecting the ground circuit to the DB25 cable . . . 10
10.3.3 Connecting the LED and sensor conductors . . . 10

11 Final Assembly **10**
11.1 Required tools . . . 10
11.2 Materials . . . 11
11.3 Procedure . . . 11
11.3.1 Positioning and shrinking the heat-shrink tubing . . . 11
11.3.2 Securing the DB25 cables to the bowls . . . 11
11.3.3 Making and attaching the feet . . . 11

Workshop Resources

Summary table of the assembly stages, required tools and materials, and estimated completion time for a group of 12 students:

https://docs.google.com/spreadsheets/d/16BShQwQICNdTbGfFbWgz5b8PJe-1twqu/edit?usp=drive_link&ouid=109005322539614617821&rtpof=true&sd=true

Playlist containing all video tutorials:

https://youtube.com/playlist?list=PL_k3PczvzYDb-f5t7U6dC4wk6cq4hSw6F&si=5ZHYeenZAPfkI9HT

1. Drilling the Bowls

Video tutorial: https://youtu.be/Srb9kXRlCo4

Required tools

- A3 printer
- Scissors
- Permanent marker suitable for smooth surfaces
- Drill
- 3–5 mm drill bit

Materials

- One or two A3 sheets
- 40 cm and 45 cm bowls
- Single-sided adhesive tape

Procedure

Preparing the templates

Print the templates on A3 paper. Double-sided printing may be used. Cut along the circular outline, then make a hole in the centre of each template.

Marking the bowls

- Place the bowls upside down on a table.
- Drill a hole in the centre of each bowl.
- Align the centre of each template with the centre of its bowl by passing the drill bit through the template's central hole and then into the bowl's central hole.

  Note: Use the ABC template for the 45 cm bowl and the B♭–D♭–E♭ template for the 40 cm bowl.

- Secure each template to its bowl at two or three points with single-sided adhesive tape. This prevents the template from rotating during marking and is very important.

- Mark the bowls 8 cm from the rim and write the note name beside each hole on the outside of the bowls.
- Drill the bowls at the marked positions. Then write the note name beside each hole on the inside, ensuring that it matches the name written outside.

2. Cutting and Piercing the Foam Discs

Video tutorial: https://youtu.be/3l_8JzbdJwk

Required tools

- Scissors
- CD
- Felt-tip pen

Materials

- Enough 1 cm-thick foam to cut twelve discs, each 12 cm in diameter
- Enough 2 cm-thick foam to cut two discs, each 12 cm in diameter

Procedure

Drawing the circles on the foam

Use the CD as a tracing template. Position it near the edge of the foam sheet, leaving an overhang of approximately 0.5–1 cm, and trace around the CD.

Cutting and piercing the discs

- Cut inside the traced line, leaving a margin of approximately 0.5–1 cm.
- Make a hole in each disc approximately 1 cm from its centre, using the tip of the scissors.

3. Attaching the Sensor and Foam Disc to the CD

Video tutorial: https://youtu.be/R5TMR4DGeH0?feature=shared

Required tools

- Scissors
- A rigid drinking-straw-type tube, made of cardboard or plastic, with a maximum diameter of 5 mm

Materials

- Piezoelectric sensors with pre-soldered wires
- Pierced foam discs
- 5 cm-wide double-sided adhesive tape

Procedure

Making the "straw" tool

Cut one end of the straw at an angle. The resulting piece should be the same length as the sensor wires.

Applying adhesive to the CD

- Cut two trapezoids from the double-sided adhesive tape. Each piece must be smaller than the CD, with approximate dimensions $L = 13$ cm, $l = 9$ cm and $h = 5$ cm.
- Apply one trapezoid to the CD, aligning the long edge of the strip with the CD's diameter so that it crosses the centre hole.
- Apply the second trapezoid beside the first so that the centre hole is covered. Do not overlap the two adhesive pieces.
- Trim any adhesive extending beyond the edge of the CD.

Attaching the sensor and foam disc

- Remove the protective films from the adhesive trapezoids attached to the CD.
- Position the sensor at the centre of the CD and press lightly to bond them together.
- Feed the sensor wires through the straw, with the angled end of the straw at the free ends of the wires and the straight end facing the sensor. Twist the wires together to make insertion easier.
- Pass the straw through the hole in the foam disc.
- Slide the foam disc towards the CD while keeping it centred, using the straw as a guide. The hole in the disc must align with the solder joints where the wires meet the sensor.
- Check that the foam disc is centred on the CD, then press the foam firmly onto the CD to bond them together.

4. Attaching the Striking Surfaces (Pads) to the Bowls

Video tutorial: https://youtu.be/PlesHf17iok?feature=shared

Required tools

- Scissors
- Cardboard to protect the work surface

Materials

- Spray adhesive or double-sided adhesive tape
- Adhesive and expanding-foam remover, such as acetone

> **Note:** Attach the pads in the following order: begin with the central note, using the pad made with 2 cm-thick foam. Then attach the peripheral pad bearing the same note name, which is the octave, using a 1 cm-thick foam pad. Before proceeding, draw a point on the edge of the central pad, directly opposite and aligned with the octave pad.

Procedure

- Twist the wires together to make it easier to feed them through the hole in the bowl.
- Cut a 2 cm piece of double-sided adhesive tape. Apply it to the pad's foam as close as possible to the wires, or apply it to the bowl as close as possible to a hole. Remove the protective film.
- Place the bowl upside down on a table.
- Feed the wires through the hole. The point where the wires exit the pad must face upwards.
- Position the pad on the bowl and press firmly.

  Alternative: When using spray adhesive, spray the adhesive around the corresponding hole in the bowl instead of using double-sided tape.

5. Making and Attaching the Supports

Video tutorial: https://youtu.be/RaQdDFHfXVI?feature=shared

Required tools

- Scissors
- Permanent marker
- Cardboard to protect the work surface

Materials

- Spray adhesive or double-sided adhesive tape
- Foam offcuts or expanding foam
- Adhesive and expanding-foam remover, such as acetone

Procedure

- Cut crescent-shaped pieces from the foam offcuts.
- Cut a 2 cm piece of double-sided adhesive tape, apply it to a foam crescent, and remove the protective film.
- Insert the foam crescent beneath the pad, with its pointed ends facing the centre of the bowl. Press it firmly against the bowl.

  Alternative: When using spray adhesive, spray adhesive onto the bowl beneath the pad at the point where the pad is furthest from the bowl, approximately 5 cm from the rim. This method is stronger but requires drying time and greater care when handling the bowl during construction. It also makes disassembly more difficult if a repair is needed.

  Alternative: The foam crescents may be replaced with expanding foam. This gives the pads better adhesion and stability, but requires additional drying time. Any excess foam must be trimmed after it has fully dried.

6. Drilling the Bowls for the LEDs

Video tutorial: https://youtu.be/Eq6pYfNYuWM?feature=shared

Required tools

- Drill
- Drill bit matching the diameter of the LEDs
- Cloth or vacuum cleaner

Materials

No additional materials are required.

Procedure

- Place the bowl upside down, with its rim resting on the table.
- Drill one hole near the edge of each CD, less than 1 cm away, on the side facing the centre of the bowl.
- Remove the drilling dust with a cloth or vacuum cleaner, then carefully remove any splinters or burrs from the holes.

7. Inserting the LEDs into the Bowls

Video tutorial: https://youtu.be/NGQcG9u3N-Y?is=k9aHxWO3TmvNvXqQ

Required tools

- Side cutters

Materials

- Transparent instant adhesive
- Adhesive remover

Procedure

- Shorten the shorter lead of each LED by 2–3 mm so that it can be identified easily.
- Apply adhesive to the LED and to the hole.
- Insert the LED into the hole with the short lead on the left.

8. Ground Circuit

Video tutorial: https://youtu.be/tKWcBon72ZY?is=V0iVByUxMFY0HdpO

Required tools

- Side cutters
- Soldering iron
- Damp sponge
- Wire stripper

- Scissors
- Utility knife
- Permanent marker
- Measuring tape

Materials

- Solder
- Heat-shrink tubing
- Ground wire

Procedure

Connecting the sensor grounds to the LEDs

- Cut fourteen 2 cm lengths of heat-shrink tubing.
- Strip the sensor ground wires if necessary.
- Bend the centre lead, or cathode, of each LED towards the ground wire of the corresponding sensor.
- Pass the central sensor's ground wire through one piece of heat-shrink tubing.
- Hold the end of the wire against the LED's centre lead, using a tool if necessary, and solder the joint.
- Repeat for every note.

Connecting the LEDs together

1. Cut a ground wire long enough to connect the cathode of the central LED to the cathode of one peripheral LED. Strip both ends.
2. Hold one end of the wire against the cathode of the central LED and solder it.
3. Prepare a wire long enough to connect the peripheral LED of the lowest peripheral note to the adjacent LED: B to D in the large bowl, and C♯ to F♯ in the small bowl. Strip both ends.
4. Hold both the wire already connected to the central note and the free wire against the cathode of the lowest peripheral note, B in the large bowl or C♯ in the small bowl, and solder them.
5. Repeat from the previous step with each LED immediately to the left until all peripheral notes have been connected. Do not reconnect the final LED to the first one.

   Note: It is preferable to wait until the testing stage before shrinking the tubing around the solder joints.

9. Preparing the DB25 Cables

Video tutorial: https://youtu.be/j8yjBHBZ7oQ?is=TlLVSO2ycR7XFWDK

Note: The tutorial uses a purpose-built DB25 cable tester designed to save time during group manufacturing workshops.

Required tools

- Multimeter or cable tester
- Four-colour pen

Materials

- Self-adhesive labels or writable adhesive tape

Procedure

Preparing the identification labels

- Write the conductor numbers and note names on the labels using the following template: https://drive.google.com/drive/folders/1wpoGsM1gzvmx8smcnHuNcGzMdusiLYeU
- Separate the labels for each bowl to avoid incorrect distribution or confusion between notes.

Cutting and stripping the cables

- Cut each DB25 cable 80 cm from the connector.
- Use a utility knife to remove 25 cm of the outer sheath, taking care not to cut the individual conductors.
- Strip every conductor.

Identifying the conductors

- Select one conductor and hold it against one multimeter probe.
- With the other probe, test each pin of the DB25 connector until the matching pin is found.
- Determine the pin number using the wiring diagram at https://docs.google.com/spreadsheets/d/1b5HFzodXUAxeUmf6LbNj5p8tK-nhCeLw/edit?usp=drivesdk&ouid=109005322539614617821&rtpof=true&sd=true, then attach the corresponding label to the conductor.
- Repeat the operation, taking care not to mix the labels for the two bowls.
- Unidentified conductors in the B♭ bowl cable may be shortened so that they do not interfere with the following stages.

10. Connecting the DB25 Cables

Video tutorial: https://youtu.be/8Uv80K08HB4?is=GaTX8PpQ3HyFEN3Q

Required tools

- Soldering iron
- Damp sponge
- Scissors

Materials

- Solder
- Heat-shrink tubing

Procedure

Preparation

- Tin all stripped conductors of the DB25 cable.
- Cut forty-two pieces of heat-shrink tubing, each approximately 1 cm long. When several tubing colours are available, prepare fourteen blue sleeves, fourteen red sleeves and fourteen sleeves in another chosen colour, referred to here as CC, which must be different from black, red and blue.

Connecting the ground circuit to the DB25 cable

- Pass the black ground conductor of the DB25 cable through the heat-shrink sleeve on the LED that terminates the ground circuit: C in the large bowl and E♭ in the small bowl.
- Align the end of the conductor with the existing solder joint and solder it in place.

Connecting the LED and sensor conductors

- Select one conductor.
- Select a sleeve matching the colour used on its label. Use the CC sleeve for labels written in black, then pass the conductor through the sleeve.
- Following the colour code below, pass the LED lead or the sensor's red wire through the same sleeve, alongside the conductor and in the same direction.
- Align the conductor with the LED lead or sensor wire and solder the joint.
- Repeat this operation for every conductor.

Colour code:

- Blue: short LED lead, positioned on the left
- Red: long LED lead, positioned on the right
- CC / black label text: red sensor wire

11. Final Assembly

Required tools

- Scissors
- Permanent marker suitable for smooth surfaces
- Drill
- 5 mm drill bit
- Hot-glue gun

- Heat source, such as a mini heat gun, lighter or soldering iron

Materials

- Hot-glue stick
- Two plastic cable ties
- Camping mat or pieces of dense foam, 5–10 mm thick
- Double-sided adhesive tape

Procedure

Positioning and shrinking the heat-shrink tubing

- After confirming that the device operates correctly by connecting it to an Arduino or ESP enclosure, position each sleeve so that it fully covers its solder joint.
- Apply heat briefly to shrink the tubing without melting the solder joints.

Securing the DB25 cables to the bowls

- Position the DB25 cable between the two lowest peripheral notes: B and C in the large bowl, or C♯ and E♭ in the small bowl.
- Mark two points, one on each side of the cable.
- Move the cable aside and drill at the marked points.
- Remove the dust with a cloth or vacuum cleaner.
- Feed the narrow end of the cable tie through one hole from the inside, then bring the same end back through the second hole from the outside.
- Position the DB25 cable between the two holes.
- Close and tighten the cable tie.
- Adjust the cable length so that approximately 5 cm of unstripped cable extends beyond the cable tie towards the centre of the bowl.
- Apply hot glue and allow it to set.

Making and attaching the feet

- Cut eight pieces measuring 5 cm by 1.5 cm from the camping mat or dense foam.
- For each foot, apply double-sided adhesive tape or hot glue to one face. Remove the protective film when using adhesive tape.
- Attach the foot to the rim of the bowl, avoiding the DB25 cable.
- Repeat until four feet have been evenly distributed around each bowl.